



JB³ ISIKOLO Ai PRODUCT SHEET

Category: On-Device Edge AI / Educational Technology (EdTech)

Status: Multi-Platform Functional Prototype

Edition: Community Edition (Open Source)

Supported Platforms: iOS & Android Cross-Platform Deployment

WHAT IS IT?

JB³ ISIKOLO AI is a state-of-the-art, multi-platform educational AI assistant designed specifically to operate under strict infrastructure constraints. By executing lightweight, highly optimized Large Language Models natively on standard commercial mobile hardware, ISIKOLO AI acts as a patient, encouraging, and brilliant local school teacher that fits right in a student's pocket.

The system comes preloaded with the full CAPS (Curriculum and Assessment Policy Statement) framework the official, centralized educational blueprint used by the South African Department of Basic Education (DBE) to guide teaching and learning from Grade R to Grade 12.

Crucially, the entire system is 100% offline, requiring zero data, zero internet access, and no remote cloud servers to deliver real-time, curriculum-aligned educational support. This platform is available completely FREE for anyone who has a compatible cell phone, computer, or device.

WHY IT MATTERS

The digital divide in emerging markets is fundamentally driven by two barriers: volatile network connectivity and prohibitive cellular data costs. Traditional AI solutions rely entirely on expensive cloud infrastructures that systematically lock out the students who need interactive tutoring the most.

ISIKOLO AI shifts the paradigm by bringing advanced cognitive computing power directly to the device edge. By running entirely on-device, it guarantees:

- **Democratic Access:** High-quality educational support accessible anywhere—from remote rural communities to deep baseline classrooms without cell towers.
- **True Zero Ongoing Costs:** Once installed on a device, the application incurs absolutely no network data overhead for the student, school, or parent.
- **Absolute Privacy & Safety:** Conversations never leave the device, ensuring total child data safety, security, and protection by default.

KEY FEATURES

- **100% Air-Gapped Operation:** Core interaction pipelines, translation matrices, and inference layers operate entirely offline, shielded from connectivity drops.
- **Preloaded CAPS Knowledge Base:** Built-in repository of the official DBE curriculum, allowing the AI to ground its answers strictly within verified local academic standards from Grade R to Grade 12.
- **Multilingual Voice Interface Layer:** Integrated native, offline Speech-to-Text (STT) and text-to-speech (TTS) synthesis modules, enabling students to interact using natural voice inputs and receive spoken local responses.
- **The "Isikolo" Pedagogical Persona:** Custom-modeled to act as an empathetic, adaptive educator who translates complex science, technology, and mathematical jargon into simple, student-friendly analogies.
- **On-Device Vector RAG Guardrails:** Operates an embedded vector database to instantly index and pull exact CAPS framework syllabus snippets, keeping the AI response factually anchored to the official classroom syllabus.

TECHNICAL SPECIFICATIONS

ISIKOLO AI bridges standard web layout components with highly specialized, bare-metal hardware acceleration wrappers.

- **Core Framework Stack:** Engineered as a decoupled hybrid application using the Capacitor runtime bridge to seamlessly map lightweight frontend user interfaces directly to a compiled native C++ background worker.
- **The Native Inference Engine:** Powered by a low-level native compilation of the Llama.cpp framework embedded inside the mobile target runtime directories.
- **Google Gemma Core Logic (E2B - Effective 2B):** Features Google's state-of-the-art open-weights model optimizing 2.3 billion active parameters (5.1B total). Extensively quantized to compact its footprint into under 1.5GB to 3GB of RAM, allowing real-time execution on typical commercial smartphones.
- **Localized MzansiLM Module:** A highly efficient 125M parameter decoder-only language model optimized on local data corpora. Operating with a hidden size of 512, intermediate size of 1536, 30 layers, 9 attention heads, and a 2048 context length, it handles localized linguistic processing and translation tasks entirely on-device.
- **Hardware Layer Acceleration:** Fully wires into native platform performance matrices—engaging Apple Silicon's Metal execution layers on iOS and the Neural Networks API (NNAPI) on Android to maximize on-chip NPU and GPU hardware capabilities.

THE FOUR PILLARS OF THE ISIKOLO FRAMEWORK

The platform's user experience is anchored around four foundational pillars designed to maximize student engagement and community value:

- **UmhloloWesifundo** (The Magic of Study): Fostering direct, curiosity-driven interactive learning where abstract concepts are transformed into immediate, understandable offline insights.
- **IsixwayisoSobalimi** (The Guide for Growers / Mentorship): Providing structured pedagogical guardrails and mental formatting that safely cultivate a student's critical thinking capacity over time.
- **IfundoYomphakathi** (Community Study): Localizing educational context

through multi-language compatibility (English, isiZulu, isiXhosa), ensuring learning honors shared social and cultural realities.

- **Imali Yesikolo (School Wealth/Capital):** Preserving community resources and family capital by completely eliminating recurring data costs and reliance on hardware server fees.

USE CASES

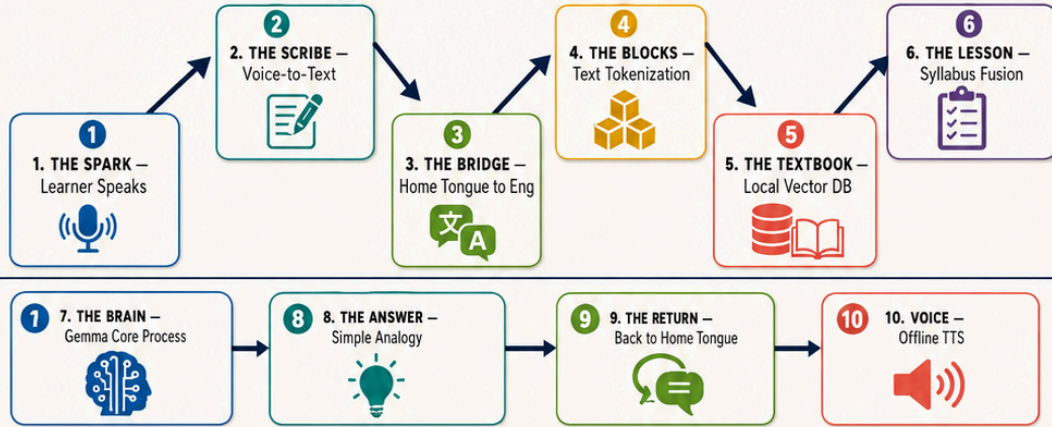
- **Data-Free Homework Support:** Learners can deeply query physics equations, history timelines, or language structures at home late at night without requiring data tokens or parent cell contracts.
- **Connectivity-Independent Classrooms:** Teachers in schools with structural cellular reception dead zones can deploy the app as an active, localized teaching assistant to generate classroom quiz structures and cross-reference syllabus rules instantly.
- **Linguistic Scaffolding:** Students can formulate complex academic thoughts in their native home languages, leveraging the offline dual-model pipeline to smoothly bridge comprehension gaps to English testing environment.



THE 10-STEP NEURAL PIPELINE: FROM CURIOSITY TO COMPREHENSION

THE NDEBELE CLASSROOM FLOW

1 10-STEP PIPELINE FLOW



2 TEN STEP CARDS

1 Learner Voice Input – The Spark of Curiosity Technical: The smartphone microphone captures raw analog audio directly from the learner. Classroom: A student asks a question aloud in isiZulu, isiXhosa, English, or another language. Edge Advantage: Fully air-gapped. No voice recordings are sent to the cloud.	2 On-Device Speech-to-Text (STT) – Writing the Spoken Word Technical: A lightweight acoustic model converts audio waves into text characters. Classroom: The phone acts like a fast classroom scribe, instantly writing down the spoken words.	3 MzansiLM Translation Layer – The Language Bridge Technical: MzansiLM translates local South African phrasing into English. Classroom: If the learner speaks in a home language, the local language helper translates it for the reasoning engine.	4 Text Tokenization – Breaking into Learning Blocks Technical: The English text is split into numeric tokens using a local vocabulary matrix. Classroom: The sentence is broken into small numbered puzzle blocks so the AI can read it piece by piece.	5 Local Vector DB Query (RAG) – Consulting the CAPS Textbook Technical: The tokens generate an embedding that queries an on-device database with official DBE guidelines. Classroom: Before answering, the AI checks its digital copy of the South African CAPS curriculum stored on the phone.
6 Context Injection – Gathering the Lesson Plan Technical: Relevant CAPS snippets are appended into the prompt memory stack. Classroom: The app stitches the learner's question together with the official lesson guidance into one guided lesson file.	7 Google Gemma 2B Processing – The Teacher's Brain Sparks Technical: The consolidated prompt is processed by Google Gemma 2B running locally on the device. Classroom: The AI teacher reads the question and CAPS rules together to work out the correct answer. Hardware Magic: On iPhone, this runs on-device via Apple Metal and the mobile GPU.	8 Educational Content Synthesis – Crafting the Perfect Analogy Technical: Gemma outputs a response stream guided by a strict pedagogical system prompt. Classroom: The answer is written like a patient teacher: simple, visual, and easy to understand.	9 MzansiLM Back-Translation – Returning to the Home Tongue Technical: The English answer is translated back into the learner's original language. Classroom: The brilliant answer comes back in the student's home language so it feels natural and familiar.	10 Offline Text-to-Speech (TTS) – The Classroom Voice Speaks Technical: An on-device speech engine converts the final text into clear audio waveforms. Classroom: The phone speaks the answer aloud in a warm local voice. The lesson is complete, fully offline.



Showcase Tip

When judges ask how a tiny phone can replace a cloud server without data, point to **Step 5** and **Step 7**. The local Vector DB and the mobile processor work together to deliver **secure, air-gapped learning**.



Offline. Secure. CAPS-aligned. Built for South African classrooms.

